Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Dayarampur, Natore-6431

CSE 19th Batch, Summer-2025
Course Code: MATH-2141
Course Title: Vector Analysis, Matrices and Laplace Transformation

1. Classify which of the following physical measurements are vectors and scalars: Volume, velocity, mass, acceleration, length, speed, temperature, momentum, force, power, pressure of gas, temperature gradient, displacement, work, potential, kinetic energy, electric charge, electric intensities, magnetic momentum.
2. Find the area of the triangle having vertices at $A(2, -3, -2)$, $B(-2, 3, 2)$, and $C(4, 3, -1)$.
3. A car is driven eastward for a distance of 5 miles, then northward for 3 miles and then in a direction 30^0 east of north for 10 miles. Draw the vector diagram and determine the total displacement of the car from its starting point.
4. What is the unit vector perpendicular to each of the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + 4\hat{j} - \hat{k}$? Calculate the sine of the angle between these vectors.
5. Find the work done in moving an object a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by $\boldsymbol{F} = 4\boldsymbol{i} - \boldsymbol{3j} + 2\boldsymbol{k}$.
6. A wagon is pulled horizontally by exerting a constant force of 12 lb on the handle at an angle of 60^0 with the horizontal. How much work is done in moving the wagon 50 ft?
7. Find the angle between the surfaces $x^2 + y^2 + z^2 = 36$ and $z = x^2 + y^2 - 4$ at the point (2, -2, 3)
8. A fluid motion is given by $\vec{v} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$. Show that the motion is irrotational and hence find the velocity potential.
9. Find the directional derivative of $\emptyset(x, y, z) = x^2yz + 4xz^2$ at (1, -2, 1) in the direction of $2\boldsymbol{i} - \boldsymbol{j} - 2\boldsymbol{k}$. Find the greatest rate of increase of $\emptyset$.
10. Let $\vec{\nabla}\phi = 2xyz^3\hat{i} + x^2z^3\hat{j} + 3x^2yz^2\hat{k}$. Find $\phi(x, y, z)$ if $\phi(1, -2, 2) = 4$.
11. Find constants a, b, c such that $\boldsymbol{v} = (x + 2y + az)\boldsymbol{i} + (bx - 3y - z)\boldsymbol{j} + (4x + cy + 2z)\boldsymbol{k}$ is irrotational. Show that $\boldsymbol{v}$ can be expressed as the gradient of a scalar function.

Md. Abdullah Al Mamun
Assistant Professor & Head
Dept. of Mathematics
BAUET